

1.3 Interface Types, Environment & Feedback

یوزر انٹرفیس وہ واسطہ ہے جس سے انسان سسٹم سے بات کرتا ہے: GUI تصاویر اور مینو، CLI لکھی کمانڈ، ٹچ، آواز (VUI)...

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 06 of 15

1.3 Interface Types, Environment & Feedback

LEARN IT

Components v–vii complete the HCI picture: the look of the controls, the place where you use them, and the signals that tell you "it worked" or "try again".

v) User interface (UI)

The **user interface** is the medium through which humans interact with computer systems.

- **Graphical User Interface (GUI):** visual elements such as windows, icons and menus (Windows OS, Android).
- **Command-Line Interface (CLI):** text-based typing (MS-DOS, Linux terminal).
- **Touch interface:** tapping, swiping, pinching (smartphones, tablets).
- **Voice User Interface (VUI):** spoken commands (smart speakers).
- **Natural User Interface (NUI):** gestures, body movements or immersive environments (VR headsets, Microsoft Kinect).

vi) Environment

The **environment** is the real-world setting where a person uses a computer system. It is shaped by:

- **Physical conditions** — an ATM outdoors in sunlight, heat, noise, rain.
- **Social conditions** — a quiet classroom, a crowded bazaar, privacy at a PIN pad.
- **Technical hardware** — slow college Wi-Fi, a cracked phone screen, an old browser.

vii) Feedback

Feedback is the signal or response given to confirm or guide actions.

System feedback

Response from the computer: a "Loading..." bar, a "Message Sent!" tick, or an error

User feedback

Input from the user to the system: tapping a "Submit" button, giving a 5-star rating.

sound.

WORKED EXAMPLE

Worked example: environment and feedback at a roadside ATM in Hyderabad.

Physical: noon sunlight washes out a dim screen — the GUI must use high contrast.

Social: people stand in a queue; the user needs a privacy shield around the PIN pad.

Technical: the printer may be out of paper, so on-screen "Take your cash" plus a beep still has to confirm the transaction.

System feedback: "Please wait", cash coming out, a printed or SMS receipt.

User feedback: the user presses Confirm, or later rates the bank's app 4 stars.

On a **school portal** during board-exam week, hundreds of students on weak Wi-Fi need a visible "Loading results..." bar. Silence after clicking "View marks" is poor system feedback.

REMEMBER IT

GUI vs CLI vs NUI: GUI = pictures you click. CLI = words you type. NUI = body you move. VUI = voice you speak. Touch = fingers on glass.

Feedback rule: **system talks back; user talks in.**

COMMON MISTAKE

Mistake: Mixing up system feedback with user feedback, or saying "environment" means only the software theme (dark mode).

Fix: A tick mark is system feedback. A 5-star rating is user feedback. Environment is the *real world* around the user — sun, noise, hardware — not the colour of the app (though a good UI will adapt to that world).

اردو میں سمجھیں

یوزر انٹرفیس وہ واسطہ ہے جس سے انسان سسٹم سے بات کرتا ہے: GUI تصاویر اور مینو، CLI لکھی کمانڈ، ٹچ، آواز (VUI) اور قدرتی اشارے (NUI). ماحول وہ حقیقی جگہ ہے جہاں سسٹم استعمال ہوتا ہے — دھوپ میں اے ٹی ایم، خاموش کلاس روم، کمزور وائی فائی۔ سسٹم فیڈبیک مشین کا جواب ہے ("بھیج دیا گیا")؛ صارف کا فیڈبیک صارف کا ان پٹ ہے (سبمٹ بٹن یا پانچ ستارے)۔

CHECK YOURSELF

1. Name five types of user interface given in the source.
2. Give one physical and one social example of environment.
3. Is a "Message Sent" tick system feedback or user feedback?

Answers: 1. GUI, CLI, Touch, VUI, NUI. 2. Physical: ATM in sunlight. Social: a quiet classroom.
3. System feedback.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Feedback is:

- a. The system telling the user what happened
- b. A student evaluation form only
- c. A foreign key
- d. A sprint retrospective only

MCQ 2. A spinner/progress bar is:

- a. Feedback that work is ongoing
- b. A primary key
- c. A sort
- d. A NAND

MCQ 3. No feedback after Save causes:

- a. Double clicks / lost trust
- b. Faster CPUs
- c. Better Big O
- d. Shorter OSI

MCQ 4. The environment includes:

- a. Noise, light, device, social setting
- b. Only CSS colours
- c. Only RAM
- d. Only Agile

MCQ 5. A GUI button should look:

- a. Clickable (affordance) and then respond
- b. Like body text
- c. Invisible
- d. Like a pie

MCQ 6. Error messages should:

- a. Say what went wrong and what to do
- b. Say ERROR 0xUnknown only
- c. Be in 4pt grey
- d. Vanish in 0.1s

Extra short questions

1. Define interface and feedback.
2. Three feedback channels.
3. Example of poor environment–UI fit.
4. What is an affordance?

Extra long question

1. Audit the college portal login: environment, UI type, three feedback events (success, fail, loading). Rewrite one bad message.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. UI: the medium of interaction. Feedback: system response the user can sense.
2. Visual (message), audio (beep), haptic (vibrate).
3. Voice-only help in a noisy canteen; tiny text on a sunny ticket machine.
4. The look that suggests how to use a control (a button looks pressable).

Long

1. Lab noise/shared PCs. GUI form. Success: "Welcome Ali". Fail: "Wrong password — try again or reset". Loading: spinner. Bad: "Error 500" → "Server busy, wait 1 min or see the lab attendant".

Next lecture: 1.4 Importance of HCI